

GTIIT Spring 2025 Calculus 1M

Workshop 14: Review

Exercise 1. Test the convergence of the improper integral

$$\int_1^{\infty} \frac{\sqrt{x}}{x^3 + 1} dx,$$

and compute its value if it converges.

Exercise 2. Determine whether the series

$$\sum_{n=2}^{\infty} \frac{(-1)^n n^2}{3^n}$$

converges or diverges, and justify your answer.

Exercise 3. Find the radius and interval of convergence of the power series

$$\sum_{n=0}^{\infty} \frac{(2n)!}{n!} x^n.$$

Exercise 4. Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable with $f(1) = 2$, $f'(1) = -1$, $g(1) = 1$, $g'(1) = 3$. Define $h(x) = f(g(x^2))$. Find the equation of the tangent line to $y = h(x)$ at $x = 1$.

Exercise 5. Determine whether the series

$$\sum_{n=1}^{\infty} \frac{n!}{2^n}$$

converges or diverges.

Exercise 6. Compute

$$\lim_{x \rightarrow 0} \frac{\tan(3x) - 3x}{x^3}.$$

Exercise 7. Solve for a, b such that the improper integral

$$\int_0^{\infty} \frac{x^a}{(1+x^b)^2} dx$$

converges. Split the integral into two, one from 0 to 1 and one from 1 to ∞

Exercise 8. Determine the convergence or divergence of the alternating series

$$\sum_{n=1}^{\infty} (-1)^{n-1} \frac{\ln n}{n}.$$

Exercise 9. Let

$$f(x) = \begin{cases} Ax^3 + B, & x \leq 2, \\ \ln(x+1) + C, & x > 2. \end{cases}$$

Find the real constants A , B , and C so that f is continuous and differentiable at $x = 2$.

Exercise 10. Find all global minima and maxima of the function

$$g(x) = x^4 - 4x^2 + 4x + 1 \quad \text{on } \mathbb{R},$$

and classify each as local or global.

Exercise 11. Determine whether each improper integral converges or diverges. If it converges, find its value:

$$\int_1^\infty \frac{\ln x}{x^2} dx.$$

Exercise 12. Compute the following one-sided limits:

$$\lim_{x \rightarrow 0^+} \frac{1 - \cos x}{x^2}, \quad \lim_{x \rightarrow 0^-} \frac{e^x - 1}{x}.$$

Exercise 13. Differentiate implicitly to find $\frac{dy}{dx}$ for

$$x^2 + xy + y^3 = \sin(xy).$$

Exercise 14. Let

$$F(x) = \int_{\ln 2}^{x^2} \frac{t^2}{\sqrt{1+t^3}} dt.$$

Find $F'(x)$ and evaluate $F'(\sqrt{3})$.

Exercise 15. Perform a partial-fraction decomposition of

$$\frac{3x+5}{(x-1)(x+2)^2}$$

and hence compute $\int \frac{3x+5}{(x-1)(x+2)^2} dx$.